Activity

Group members:

Zero-Inflated Poisson (ZIP) models

In this class activity, we work with survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. The survey asks students "How many alcoholic drinks did you consume last weekend?" The data includes the following variables:

- `drinks`: the number of drinks the student reports consuming
- `sex`: an indicator for whether the student identifies as male (1 = male)
- `OffCampus`: an indicator for whether the student lives off campus (1 = off campus, 0 = on campus)
- `FirstYear`: an indicator for whether the student is a first-year student (1 = first year, 0 = not first year)

The model

To account for excess zeros in the data, we fit a *zero-inflated Poisson* (ZIP) model. Here is the probability function for our model:

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

The fitted model estimates α_i with a logistic regression component, and λ_i with a Poisson regression component. The fitted model is

$\log\left(\dfrac{\hat{\alpha}_i}{1-\hat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$

$\log(\hat{\lambda}_i) = 0.75 + 0.42 \ OffCampus_i + 1.02 \ Male_i$

Questions

1. What is the estimated probability that a first year student never drinks?

2. What is the estimated average number of drinks for a male student who lives off campus and sometimes drinks?

3. What is the estimated probability that a male first year student who lives off campus had at least one drink last weekend?